

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0473

Brevard 230 kV
September 17, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0473. The Interconnection Customer proposes a 150.9 MW solar photovoltaic facility interconnecting at the Brevard 230 kV substation in the Illinois area. The request was received on May 06, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 247 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$590,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GRAND RIDGE BATTERY PROJECTS to BREVARD 230 kV	BREVARD to GRAND RIDGE BATTERY PROJECTS SUBSTATION 230 kV	111.8	623	77.2
TROY ENERGY LLC to HIGHGATE 230 kV	MASON ST 500 kV/230 kV TX	102.2	642	7.9
GRAND RIDGE BATTERY PROJECTS to BREVARD 230 kV L	GRAND RIDGE BATTERY PROJECTS to BREVARD 230 kV Cir. 1	100.7	692	77.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GRAND RIDGE BATTERY PROJECTS to BREVARD 230 kV	Rebuild line with higher-capacity conductor	\$854,000	41% (\$347,000)
HIGHGATE to TROY ENERGY LLC 230 kV	Reconductor line	\$8,766,000	2% (\$144,000)
GRAND RIDGE BATTERY PROJECTS to BREVARD 230 kV L	Reconductor line	\$242,000	41% (\$99,000)
	Total allocated to project		\$590,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.